

DIGITAL TECHNOLOGIES

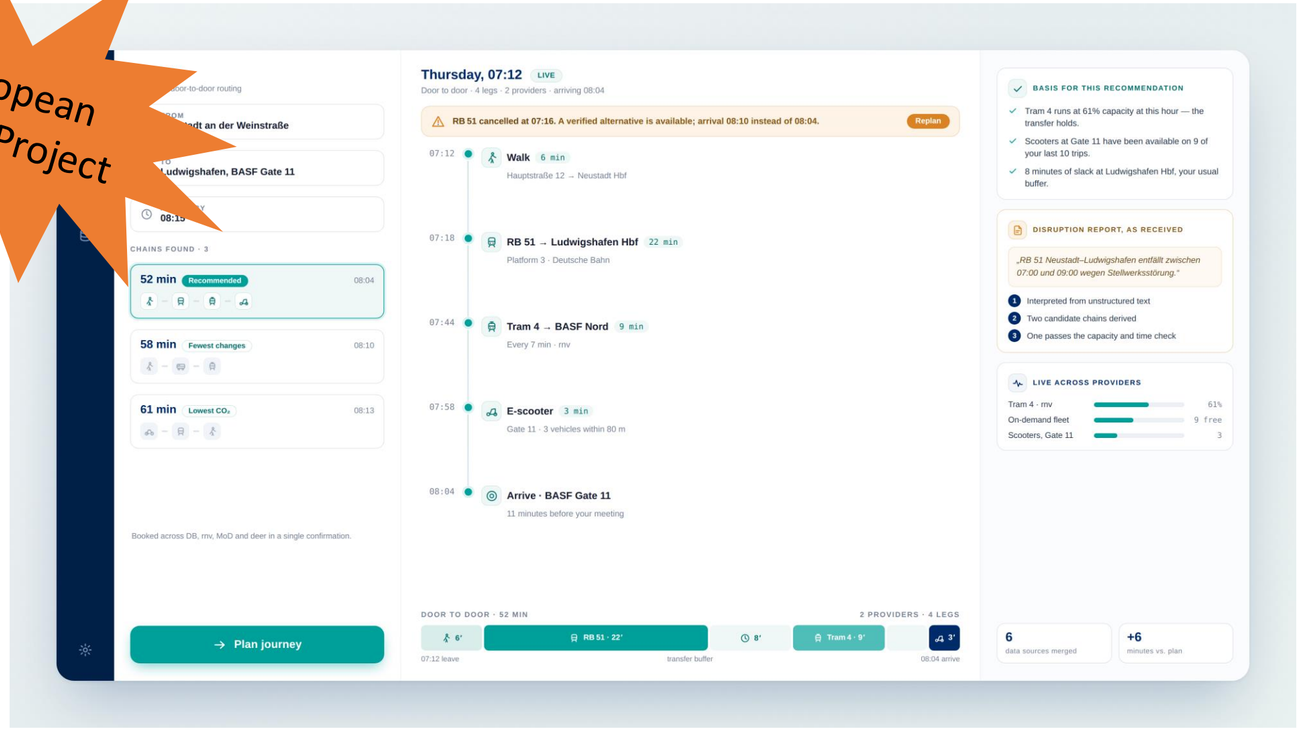
ZUKUNFT. DIGITAL. NACHHALTIG.

WiSe 2026

European Team Project

FastMile

Unified Mobility Platform for LLM-Guided Routing



Machine Learning and COgnitive SoftwaRE (CORE)
Research Group
AI Team Projects (ATP)

Ein gemeinsames Studienprogramm der

Ostfalia Hochschule für angewandte Wissenschaften

TU Clausthal

CORE

UNIVERSITATEA BABEŞ-BOLYAI
BABEŞ-BOLYAI TUDOMÁNYEGYETEM
BABEŞ-BOLYAI UNIVERSITÄT
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TRADITIO ET EXCELLENTIA

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THE STARTING POINT

Daily commuting, across every mode that fits

A polycentric region

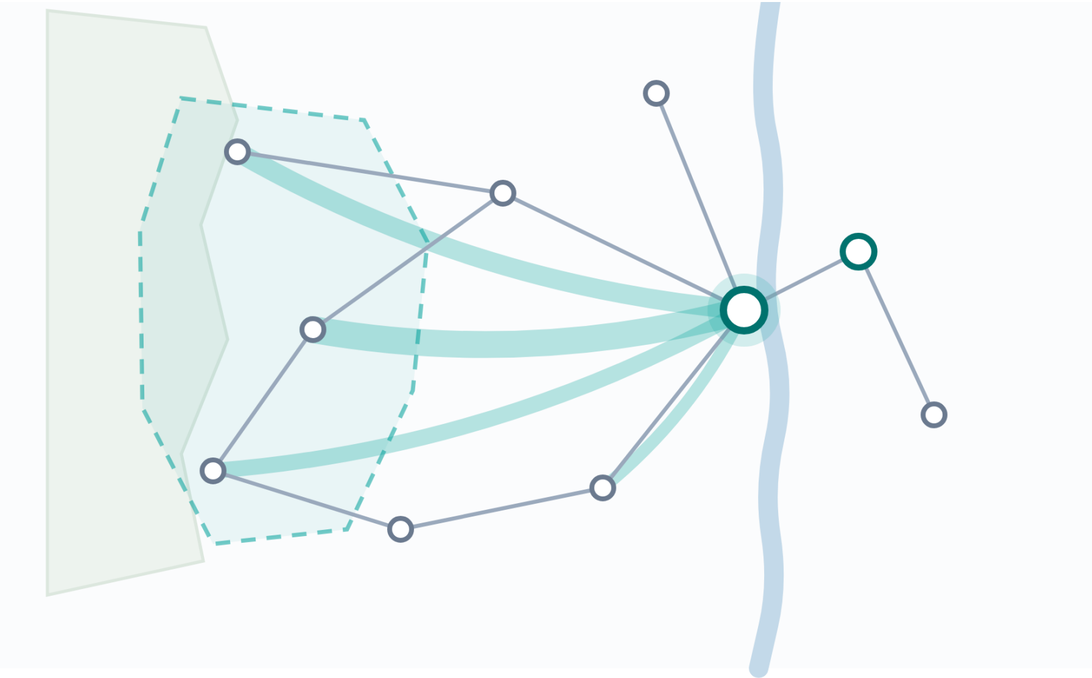
Commuters travel from surrounding towns and districts into larger employment centres

Several modes, one journey

Public transport, on-demand services, car sharing and micromobility are combined into a single door-to-door chain

The whole chain decides

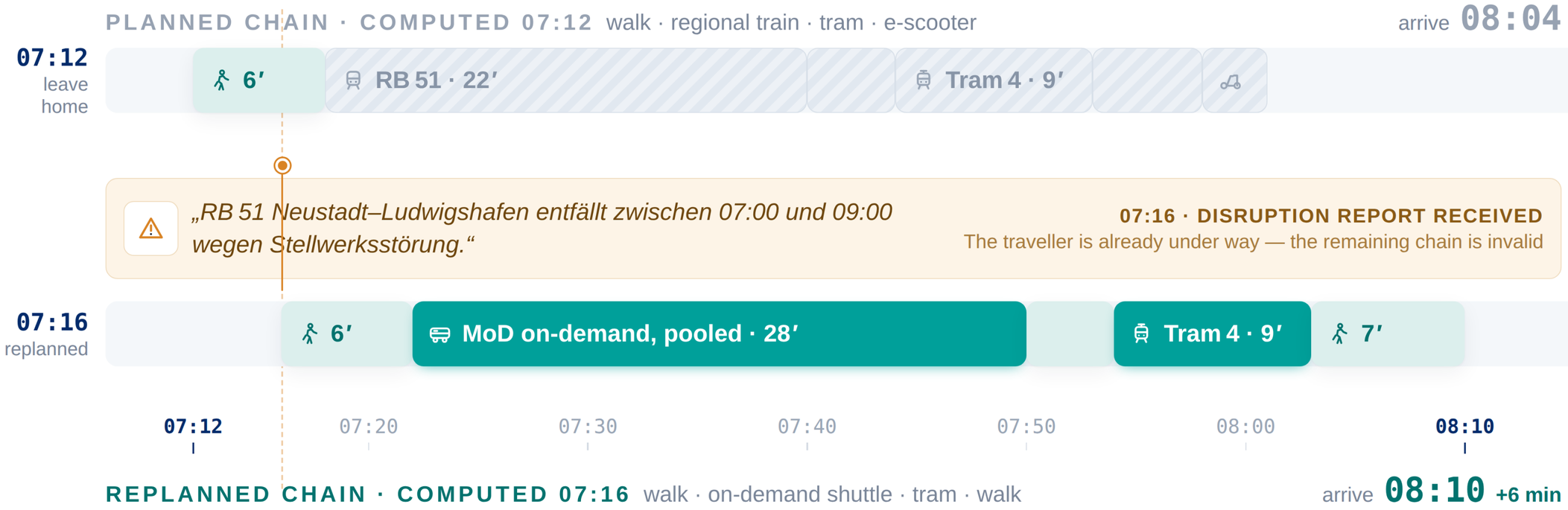
The private car still dominates because it is more often the option that works reliably from the front door to the destination



OBJECTIVE Provide convenient and predictable transfer options across the whole intermodal journey

INCIDENT HANDLING

Disruptions need to be adressed on the go



Detection
The report is read from unstructured text

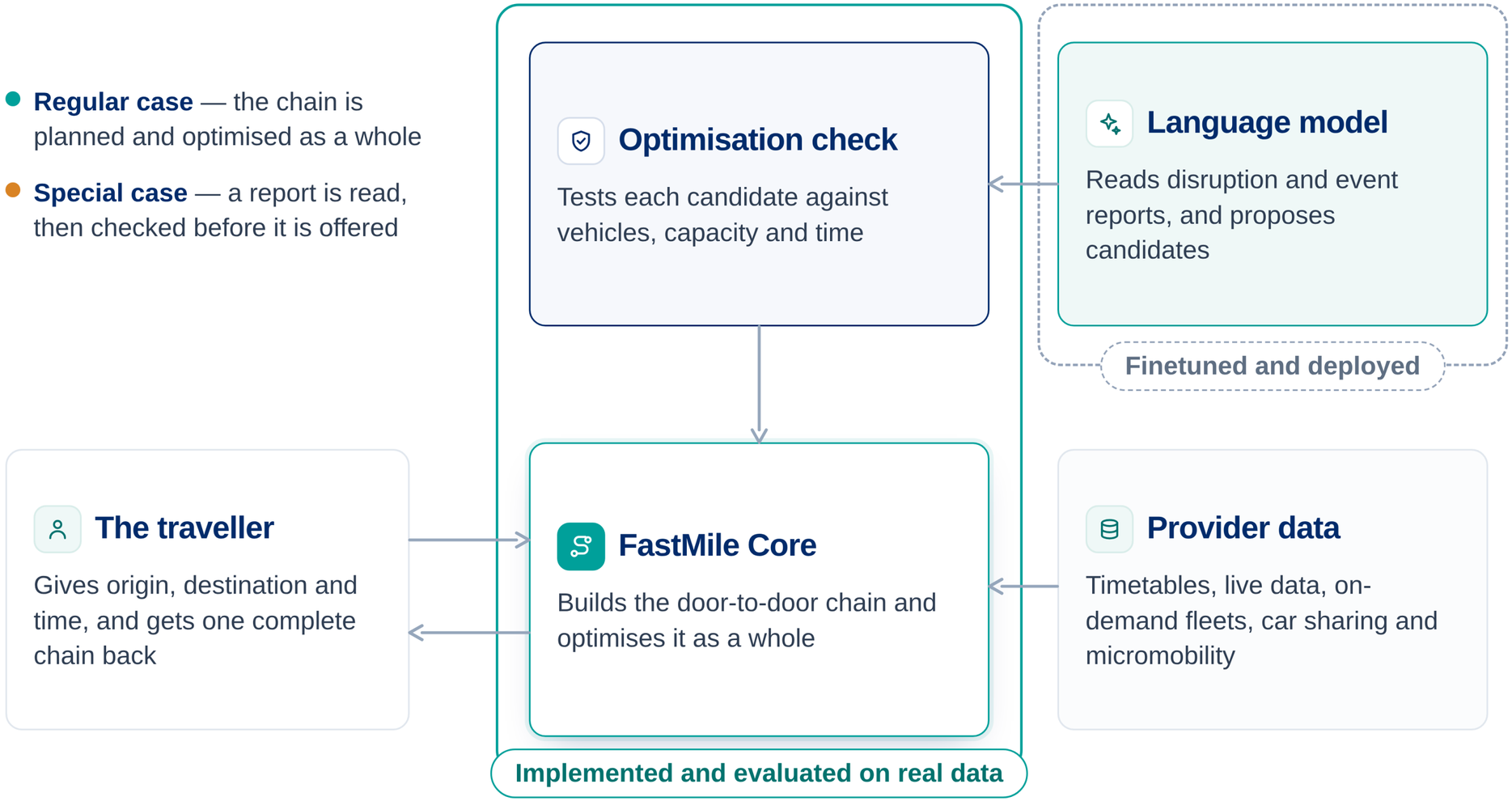
Replanning
Candidate chains are derived

Verification
Checked against capacity and remaining time

SYSTEM ARCHITECTURE

What you will implement

- **Regular case** — the chain is planned and optimised as a whole
- **Special case** — a report is read, then checked before it is offered



DATA BASIS

Data sources

SOURCE	USED FOR
Public transport timetables, stops and real-time data	Intermodal connections that work in practice
Utilisation of lines and vehicles	Capacity-aware recommendations, including at peak times
Trip data from on-demand services	Fleet state, pooling opportunities, demand patterns
Car-sharing and micromobility offers	Further modes available inside a single chain
Disruption reports and event announcements	The trigger for the special cases, arriving as free text

We use real usage data – not synthetic data

DEV-TEAM

Thinking about joining?

WHAT YOU BRING

- Comfortable writing **Python**
- Comfortable with **git** and a shared repository
- Happy working in an international team across two universities
- Interested in optimization and machine learning on live data

WHAT YOU LEARN

- Interpreting unstructured operational text with language models
- Combining machine learning with analytical route optimisation
- Building a web application on live data from several providers
- Taking a research question through to a system that runs

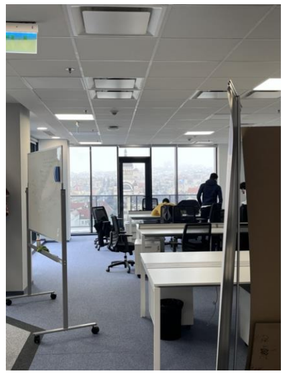
Be part of an international team!

together with students from Babeș-Bolyai University, Cluj, Romania

For more info and videos visit:

https://www.core-network.ai/ai-team-projects

Intensive Week in Cluj






Social Events





SAVE THE DATE

Intensive Week Cluj:

~ 11.11. – 19.11.2026

Costs for flight and accommodation will be covered

Intensive Week Goslar:

~ 01.02. – 07.02.2027

CORE Projects in WiSe26

Kairos	**FastMile**

Machine Learning and COgnitive SoftwaRE (CORE)

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